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School of Clinical Medicine
Department of Medicine
香港大學內科學系



瑪麗醫院
QUEEN MARY HOSPITAL

Seizure and loss of consciousness: Delirium and encephalopathy; epilepsy; coma and brain death; care of unconscious patients; electrophysiology I

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Learning objectives

- **Consciousness and it's disorders**

- Describe the differential diagnosis and management of altered state of consciousness.
- Perform the Glasgow Coma Scale (GCS).
- Identify features that differentiate syncope and seizure.

- **Seizure and epilepsy**

- Name the different classifications of seizure.
- Explain the definition of seizure and epilepsy.
- Describe the epidemiology, etiologies, and clinical features of common types of seizure/ epilepsy.
- Describe commonly used antiseizure medications and their side effects.
- Learn the emergency management of a seizure attack and status epilepticus.

- **Electrophysiology I**

- Recognize the value of EEG as an investigative tool in seizure/ epilepsy.

- **Brain death**

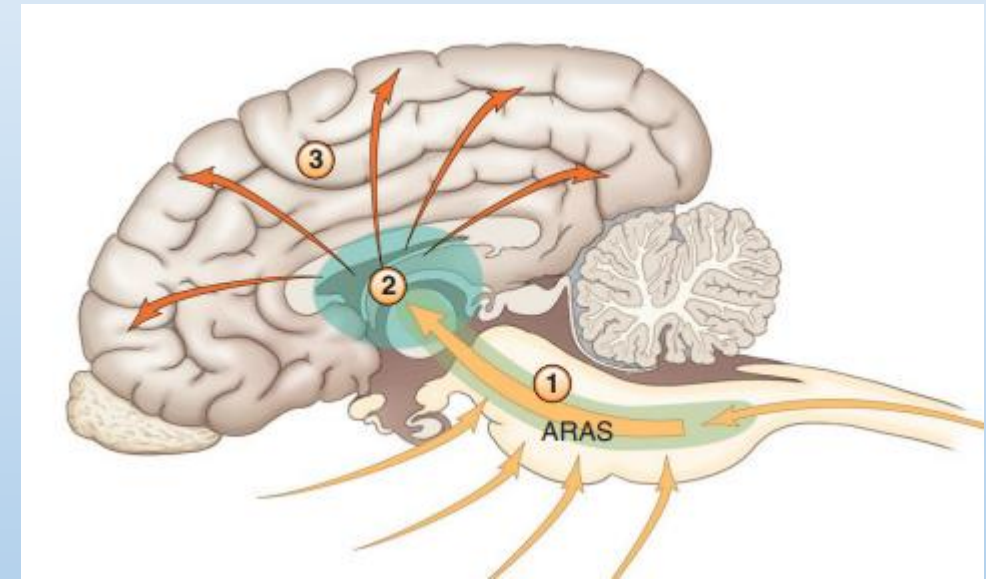
- Explain what is brain death and how it is diagnosed.

Consciousness

- State of awareness of self and environment
- **Arousal**
 - Awoken state. Primitive set of responses
 - Brainstem function (reticular activating system)
- **Awareness**
 - High level integration of multiple sensory input
 - Resides in the cerebral cortex

Reticular formation

- Complex network of brainstem nuclei and neurons that spans the entire brainstem
 - **Ascending reticular activating system**
 - Arousal/ consciousness
 - Descending reticular system
 - Muscle tone
 - Autonomic nervous system/ visceral function
- **Arousal/ Consciousness**
 - Projection to thalamus
 - Thalamic relay neurons activate cortical pyramidal neurons
 - Tonic mode (wakefulness or REM-sleep)
 - Burst mode (non-REM sleep)



A same condition, e.g. stroke, can results in any of these states depending on the location and severity

Disorder of consciousness

Condition	Awareness	Sleep-wake cycle	Motor behavior characteristic
Coma	No	No	No purposeful behavior Posturing to pain
Vegetative state	No	Yes	No purposeful behavior Eye opening Random vocalization
Minimal conscious state	Partial/ fluctuating	Yes	Inconsistent Reproducible purposeful behavior: localization to pain/ visual pursuit
Lock-in-syndrome (mimics disorder of consciousness) - Damage of descending motor tracts at the brainstem	Yes	Yes	Yes, but limited to ocular movement

Altered mental state/ delirium: Any acute change in attention/ awareness or cognition

Causes of altered state of consciousness/ coma

- Structural brain disease (damage to reticular formation OR sizable/ diffuse lesion)
 - Stroke (Ischemic/ hemorrhagic)
 - Traumatic brain injury/ epidural/ subdural hematoma
 - Brain mass/ tumor
 - CNS infection
 - Inflammation (autoimmune encephalitis)
 - Hydrocephalus

Causes of altered state of consciousness/ coma

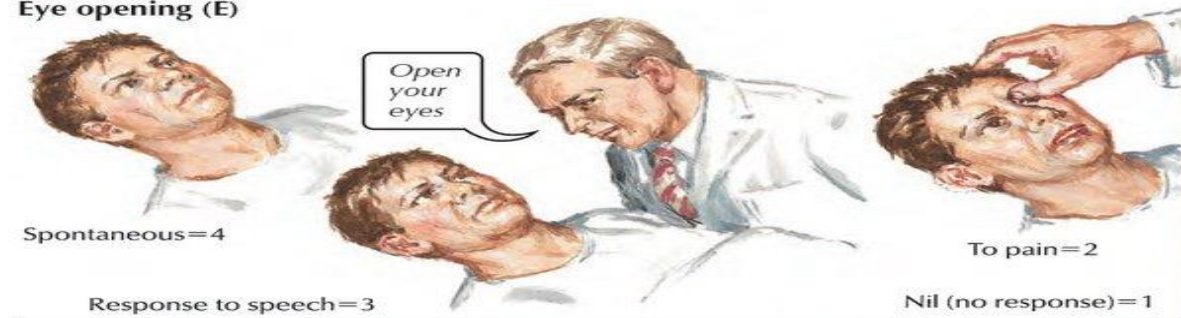
- Diffuse neuronal dysfunction
 - **Encephalopathy (group of condition that lead to brain dysfunction)**
 - Sepsis
 - Electrolyte disturbance (hyponatremia, hypercalcemia)
 - Hypo/ hyperglycemia
 - Renal failure (uremic encephalopathy)
 - Liver failure (hepatic encephalopathy)
 - Wernicke encephalopathy
 - Endocrine cause (adrenal crisis, thyroid storm, myxedema coma)
 - Toxic (alcohol, substance abuse)
 - **Seizure/ Status epilepticus (electrical abnormalities)**
 - **Cerebral hypoperfusion/ hypoxia**
 - Syncope
 - Shock

Rapid assessment

- History to identify potential cause
- PE
 - ABC
 - GCS
 - Pupillary reflexes
 - Obvious focal weakness
 - Plantar responses
 - Any trauma
 - Fever
 - Neck stiffness

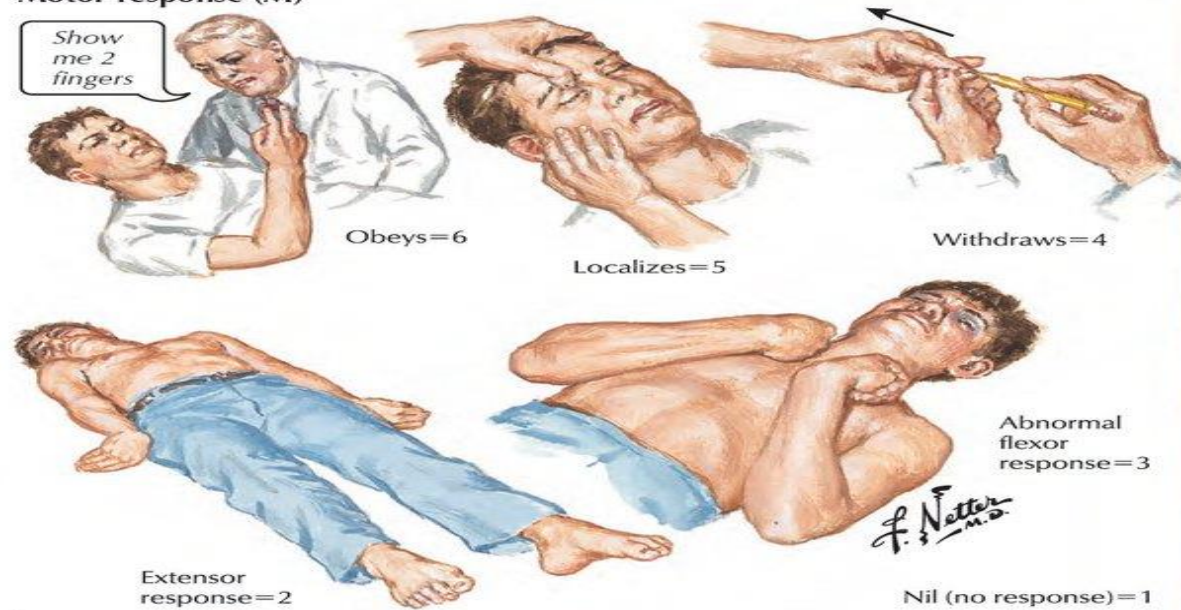
Glasgow Coma Scale

Eye opening (E)



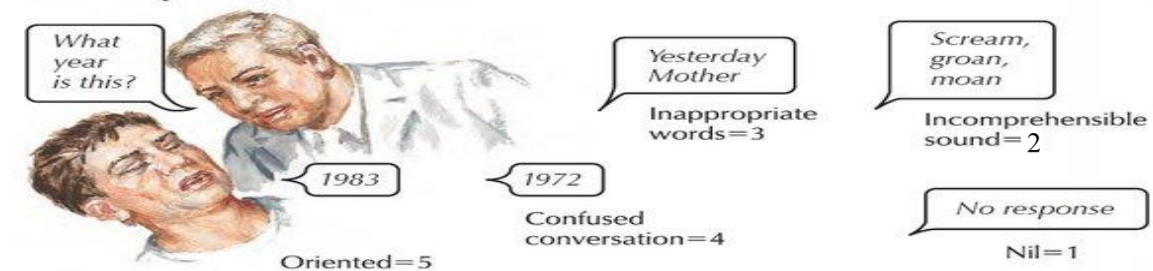
E	
Spontaneous	4
To speech	3
To pain	2
Nil	1

Motor response (M)



M	
Obeys	6
Localized	5
Withdraws	4
Abnormal flexion	3
Extensor response	2
Nil	1

Verbal response (V)



V	
Oriented	5
Confused conversation	4
Inappropriate words	3
Incomprehensible sounds	2
Nil	1

Coma score (E+M+V)=3 to 15

Clinical tips for painful stimulus

Trapezius squeeze

Supraorbital pressure



Glasgow Coma Scale

13-15: Mild

9-12 : Moderate

3-8 : Severe

Investigations

- Blood/ urine test includes routines, glucose, toxins/ drug screen
- Neuroimaging
- Based on clinical suspicion
 - Lumbar puncture
 - EEG

Management

- Supportive
 - ABC, intubation if indicated especially GCS <8 and cannot protect airway
- Specific management depend on the cause identified

Diagnosis of reversible causes of coma

Jonathan A Edlow, Alejandro Rabinstein, Stephen J Traub, Eelco F M Wijdicks

Lancet 2014; 384: 2064-76

Brain mass	Surgical removal, possible steroids (tumour, abscess)
Anoxic-hypoxic brain disease with return of spontaneous circulation after cardiac arrest	Therapeutic mild hypothermia (or at least strict maintenance of normothermia)
Raised intracranial pressure	Elevate head of bed, intravenous mannitol, hypertonic saline, hyperventilation and corticosteroids, loosen any constricting bandages or collars
Subdural or epidural haematoma	Consider surgical drainage
Intracerebral haemorrhage	Hemostatic therapy to correct coagulopathy, consider blood pressure control and possible surgical drainage
Acute ischaemic stroke	Thrombolytic therapy
Hydrocephalus	Ventriculostomy and drainage
Cerebral oedema resulting from stroke	Decompressive craniectomy
Cerebellar oedema resulting from stroke	Suboccipital decompressive craniectomy
Cerebral venous sinus thrombosis	Intravenous heparin
Sepsis	Intravenous antimicrobials; surgical drainage of any abscess; goal-directed Intravenous fluids

CNS infections	Antimicrobials, drainage for brain abscess, consider steroids for meningitis
Non-convulsive or minimally convulsive status epilepticus	Anti-epileptic drugs

Hypoglycaemia	50% dextrose
Hyperglycaemia, diabetic or alcoholic ketoacidosis	Intravenous saline, insulin
Hyponatraemia	Hypertonic saline
Hypercalcaemia	Intravenous saline, furosemide, other drugs
Hyperammonaemia	Treat underlying condition
Renal failure	Dialysis
Hepatic encephalopathy	Lactulose
Thyroid storm	Anti-thyroid medications and β -blockers
Myxoedema coma	Hormone replacement
Adrenal crisis	Hormone replacement and intravenous fluids
Pituitary apoplexy	Possible surgery, hormone replacement
Wernicke's encephalopathy	Intravenous thiamine
Sedative hypnotic agents	Supportive care
Opioids	Naloxone

Transient loss of consciousness

- Loss of consciousness with complete recovery
- Syncope (cardio or cerebrovascular cause) vs. Seizure
- History from witness is vital
 - EEG not specific and sensitive
- LOC + muscle jerk does not mean seizure
 - Syncopal myoclonus occurs in up to 20% of patients with syncope
- Misdiagnosing seizure and missing cardiovascular syncope
 - Mortality of cardiovascular syncope is much HIGHER than seizure
 - Misdiagnosing seizure -> missing cardiovascular syncope -> high mortality

Causes of syncope

Classification	Remarks
Neurocardiogenic (most common)	Increase vagal tone - Vasovagal syncope, situational syncope (micturition/defecation)
Cardiac (most lethal)	Myocardial infarction, Arrhythmia, valvular heart disease, cardiac tamponade, pulmonary embolism
Neurological	Seizure Others (less common): posterior circulation TIA, subclavian steal, complex migraine
Orthostatic	Drop in systolic BP >20mmHg on standing - Drug, volume loss, autonomic dysfunction
Unknown	Unexplained Need to rule-out potential life-threatening causes



Clinical features	Cardiogenic Syncope	Seizure
Preceding light-headedness/sweating	++	-
Chest pain/ Palpitation	+	+/-
Situation	Crowded, hot place Micturition/ defecation	Flashing lights
Postural change	++	-
Duration	Seconds to minutes	Minutes
Limb twitching	After LOC, few seconds	Occurs with LOC, at least 30-60 seconds
Urinary incontinence	+	+
Tongue or lip injury	+ through injury	++
Post event	Drowsy but not confused Orientated within minutes	Drowsiness/Confusion up to hours
Family history	+	+/-
1 year mortality	High	Low



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Seizure and epilepsy

Definition

- **Seizure**

- Sudden burst of electrical activity in the brain leading to changes in movement, behavior, feeling and/or consciousness
- Diagnosis of seizure
 - **CAREFUL HISTORY IS MOST IMPORTANT**
 - Suggestive features: transient occurrence of
 - Abnormal behavior
 - Involuntary movements
 - Recurrent/ Stereotypical attack

- **Epilepsy**

- Brain disorder that causes recurring, unprovoked seizures

Pathophysiology

- Abnormal synchronous brain activity caused by abnormal spread & neuronal recruitment secondary to
 - Enhanced excitatory transmission
 - Failure of inhibitory mechanism
 - Changes in intrinsic neuronal properties

Conditions that mimics seizure

Panic attack/ Hyperventilation	Abrupt onset intense feeling of dread/ fear; sense of impending death or inability to breath; prominent autonomic features (sweating, tachycardia, nausea); External trigger; longer (5-30 mins) than typical seizure; May have hyperventilation -> perioral/ hand numbness; carpopedal spasm
Psychogenic non-epileptic seizure	Psychiatric history; patient usually motionless with eye closed at onset; fluttering eye movements and forceful eye closure; out-of-phase, thrashing limb movement and pelvic thrusting; urinary incontinence unusual
Syncope	As previous slides
Transient global amnesia	Prominent anterograde amnesia; prolonged duration (hours); no altered consciousness, confusion, weakness or speech problems; repeated questioning
Transient ischemic attack	Negative symptoms (weakness, loss of sensation, aphasia) predominate
Migraine aura	Both positive and negative symptoms (mostly visual and sensory); last longer, slow spread of positive symptoms followed by negative symptoms; Develop headache after aura



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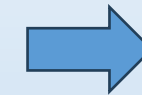
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Classification of seizure

- Provoked vs. unprovoked
- Focal vs. generalized
- Etiology

Provoked vs. unprovoked seizure

- **Provoked seizure** (acute symptomatic seizure):
Seizure due to acute focal/ diffuse brain insult
 - Not epileptic
 - Residual brain injury may subsequently cause unprovoked seizure (post-stroke epilepsy)
- **Unprovoked seizure**: Seizure occurring without an acute precipitating factor
 - Can occur due to static injury (old stroke) or progressing injury (neurodegenerative disease)



VINDICATE	
V	Vascular
I	Infection, inflammation
N	Neoplasms
D	Drugs (toxins), degenerative
I	Iatrogenic, idiopathic
C	Congenital/inherited/developmental
A	Autoimmune, allergy, anatomic
T	Trauma
E	Endocrine/metabolic, Environment/work exposure

ILAE 2017 Classification of Seizure Types Expanded Version ¹

Seizure originate in a localized foci

Begins simultaneously in both brain hemispheres

Focal Onset

Aware

Impaired
Awareness

Motor Onset

automatisms
atonic ²
clonic
epileptic spasms ²
hyperkinetic
myoclonic
tonic

Nonmotor Onset

autonomic
behavior arrest
cognitive
emotional
sensory

focal to bilateral tonic-clonic

Generalized Onset

Motor

tonic-clonic
clonic
tonic
myoclonic
myoclonic-tonic-clonic
myoclonic-atonic
atonic
epileptic spasms

Nonmotor (absence)

typical
atypical
myoclonic
eyelid myoclonia

Unknown Onset

Motor

tonic-clonic
epileptic spasms
Nonmotor
behavior arrest

Unclassified ³

Presenting symptoms

- **Focal aware seizure (simple partial seizure)**
 - Symptoms depend on the location involved (**Aura for Focal impaired awareness seizures/ generalized seizure**)
 - **Frontal**
 - **Focal motor seizure**; olfactory hallucination, aphasia, hyperkinetic seizure
 - **Temporal**
 - **Autonomic (epigastric sensation), cognition (deja/ jamais vu), emotional (fear), auditory hallucination**
 - **Parietal**
 - Somatosensory seizure, visual illusion
 - **Occipital**
 - Visual seizures (flashes/ multicolored shapes; loss of visual field)

Presenting symptoms

- Focal impaired awareness seizures (complex partial seizure)
- Seizure semiology also depend on location involved
 - Automatism, abnormal posturing, head/ eye version
 - Impaired awareness
- Classical example: temporal lobe epilepsy due to hippocampal sclerosis
 - Autonomic (epigastric sensation), cognition (deja/ jamais vu) or emotional (fear) seizures followed by behavioral arrest with progressive impaired awareness + automatism (chewing, lip-smacking, swallowing, tongue movement)/ mannerism
- Focal seizures can spread to bilateral tonic-clonic seizure (secondary generalization)

Presenting symptoms

- Generalized onset seizure
 - Motor
 - Tonic-clonic
 - Tonic
 - Myoclonic
 - Non-motor
 - Behavioral arrest/ impaired consciousness

Clinical context

- A 66/M presenting with acute right frontal stroke and recurrent left arm involuntary movement
 - Provoked seizure (acute injury); focal motor seizure
- A 25/M with recurrent epigastric sensation/ déjà vu followed by behavioral arrest. MRI brain showing hippocampal sclerosis
 - Unprovoked seizure; focal impaired awareness seizure
- A 18/F with a history of recurrent jerky movement in the morning, presenting with generalized tonic-clonic seizure. MRI normal.
 - Unprovoked seizure; myoclonic seizures + generalized tonic-clonic seizure
- A 82/F with known dementia presenting with generalized tonic-clonic seizure
 - Unprovoked seizure (progressing injury); generalized tonic-clonic seizure

Epilepsy

- At least two unprovoked seizures occurring >24 hours apart
- One unprovoked seizure and >60% chance of recurrence in the next 10 years
 - Abnormal epileptiform discharges on EEG
 - Structural brain pathology
 - Abnormality neurological examination beyond postictal period, including focal findings (presumably due to a lesion in the brain) or intellectual disability
 - Nocturnal seizure

Epidemiology

- **Incidence**

- Two peaks:

- Childhood: Genetic/ polygenetic/ unknown
 - ≥ 65 years: Structural (including neurodegenerative diseases)

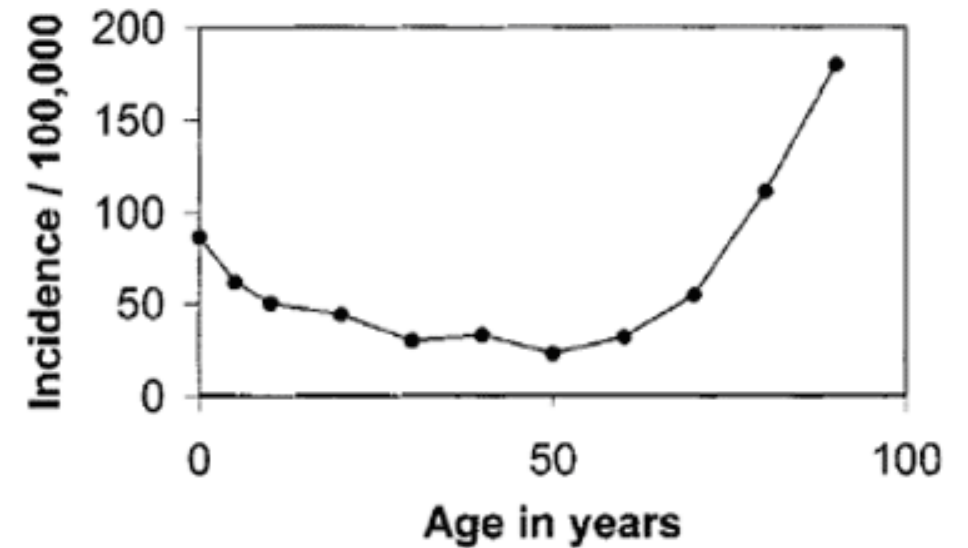
- **Prevalence**

- ~1% in adults
 - In ≥ 65 years: 2-6%

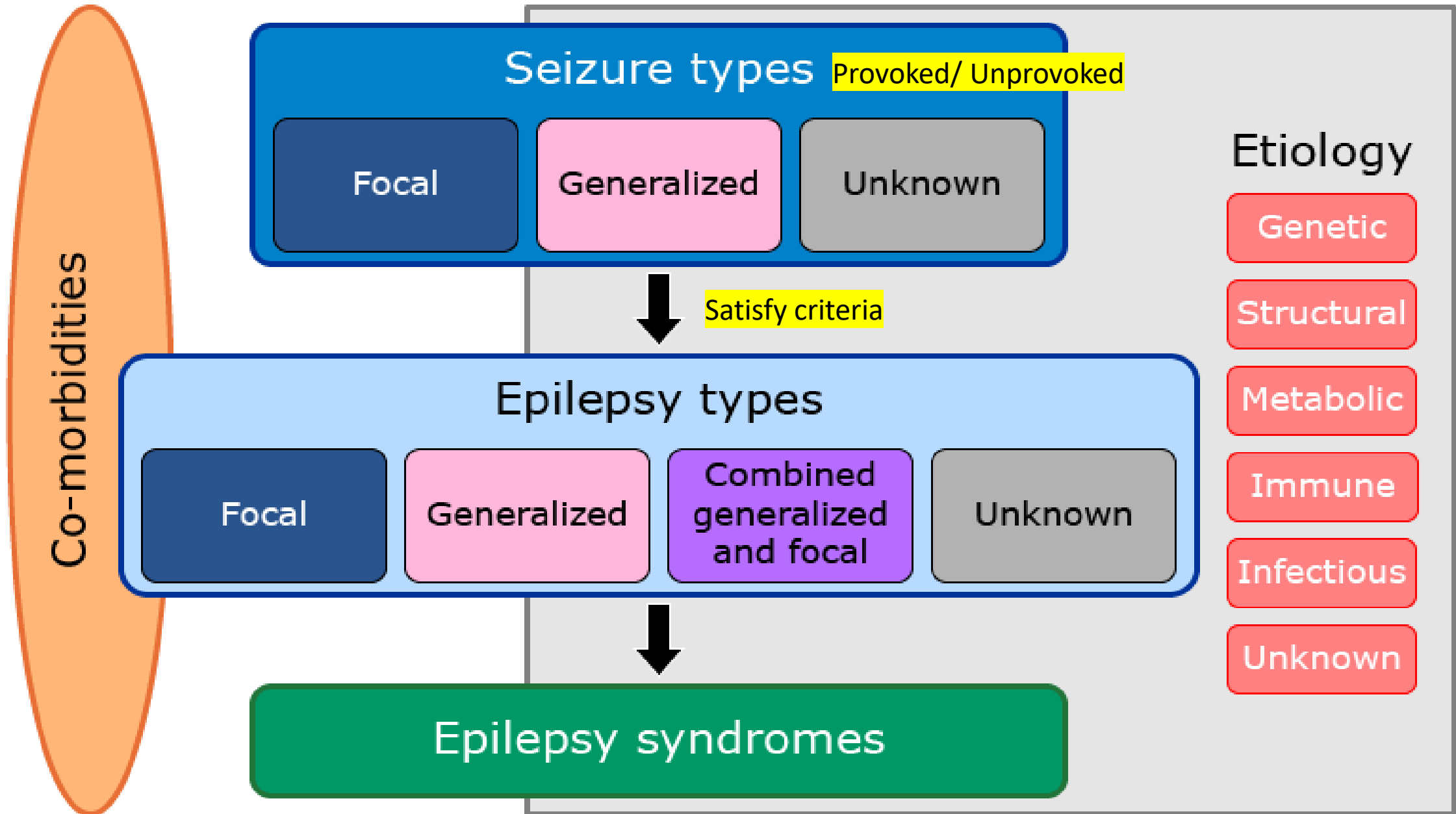
Medscape®

www.medscape.com

Age-Related Incidence of Epilepsy



International League Against Epilepsy (ILAE) framework for seizure and epilepsy classification



Etiology of seizure/ epilepsy

- **Genetic**
 - Congenital/inherited
- **Structural**
 - Vascular, Neoplasm, Trauma, Degeneration, Developmental
- **Metabolic**
 - Drug, Endocrine/ Metabolic
- **Immune**
 - Inflammation, Autoimmune
- **Infectious**
 - Infectious
- **Unknown**
 - Idiopathic

VINDICATE

- V** Vascular
- I** Infection, inflammation
- N** Neoplasms
- D** Drugs (toxins), degenerative
- I** Iatrogenic, idiopathic
- C** Congenital/inherited/developmental
- A** Autoimmune, allergy, anatomic
- T** Trauma
- E** Endocrine/metabolic, Environment/work exposure

Genetic causes

- Genetic mutation in genes encoding
 - Ion channel
 - Neurotransmitter receptors
 - Synaptic support proteins
 - mTOR pathway regulators
 - Chromatin remodeling & transcription regulators



Genetic	Chromosomal abnormalities: Down syndrome, Angelman syndrome, etc. Gene abnormalities that cause epilepsy: SLC2A1, KCNQ2, etc.
Structural	Developmental: Focal cortical dysplasia, tuberous sclerosis, grey matter heterotopia <i>Hippocampal sclerosis</i> Vascular: Stroke, vascular malformations (AVM, angioma), hypoxic-ischemic injury Neoplasm: dysembryoplastic neuroepithelial tumor (DNET), primary/ secondary brain tumours Trauma: Traumatic brain injury Degeneration: Alzheimer's disease, Parkinson disease/ Lewy body dementia
Metabolic	Electrolyte disturbance: hyponatremia, hypernatremia, hypocalcemia, hypomagnesemia Metabolic: Hypo/ hyperglycemia, Uremia, Hyperammonemia Drugs: Tramadol, Carbapenem/ Cefepime antibiotics Substance abuse/ alcohol
Immune	Autoimmune encephalitis
Infectious	CNS infection
Unknown	-

Epilepsy syndrome

- Epilepsy with **CHARACTERISTIC** clinical and EEG features, that are often supported by specific etiology

Variable Age

- Idiopathic Generalised Epilepsy Syndromes
 - Juvenile Myoclonic Epilepsy (JME)
 - Juvenile Absence Epilepsy (JAE)
 - Epilepsy with Generalised Tonic-Clonic Seizures Alone (EGTCA)
- Focal Epilepsy Syndromes
 - Sleep-Related Hypermotor (Hyperkinetic) Epilepsy (SHE)
 - Familial Mesial Temporal Lobe Epilepsy (FMTLE)
 - Epilepsy with Auditory Features (EAF)
 - Familial Focal Epilepsy with Variable Foci (FFEVF)
- Combined Generalised and Focal Epilepsy Syndrome
 - Epilepsy with Reading-Induced Seizures (EwRIS)
- With Developmental and/or Epileptic Encephalopathy (DEE) or Progressive Neurological Deterioration
 - Progressive Myoclonus Epilepsies (PME)
 - Rasmussen Syndrome

Mesial temporal lobe epilepsy with hippocampal sclerosis

Neonate/Infant

- Self-limited
 - Self-limited (Familial) Neonatal Epilepsy (SeLNE)
 - Self-limited Familial Neonatal-Infantile Epilepsy (SeLFNIE)
 - Self-limited (Familial) Infantile Epilepsy (SeLIE)
 - Genetic Epilepsy with Febrile Seizures Plus (GEFS+) Spectrum
 - Myoclonic Epilepsy in Infancy (MEI)
- With Developmental and Epileptic Encephalopathy (DEE)
 - Early-Infantile Developmental and Epileptic Encephalopathy (EIDEE)
 - Epilepsy of Infancy with Migrating Focal Seizures (EIMFS)
 - Infantile Epileptic Spasms Syndrome (IESS)
 - Dravet Syndrome (DS)

Childhood

- Self-limited Focal Epilepsy Syndromes (SeLFEs)
 - Remission expected in all cases by adolescence
 - Self-Limited Epilepsy with Centrottemporal Spikes (SeLECTS)
 - Self-Limited Epilepsy with Autonomic Seizures (SeLEAS)
 - Remission expected in most cases by adolescence
 - Childhood Occipital Visual Epilepsy (COVE)
 - Photosensitive Occipital Lobe Epilepsy (POLE)
- Genetic Generalised Epilepsy Syndromes
 - Epilepsy with Eyelid Myoclonia (EEM)
 - Epilepsy with Myoclonic Absence (EMA)
- Idiopathic Generalised Epilepsy Syndrome
 - Childhood Absence Epilepsy (CAE)
- With Developmental and/or Epileptic Encephalopathy (DE, EE or DEE)
 - Epilepsy with Myoclonic Atonic Seizures (EMaTS)
 - Lennox-Gastaut Syndrome (LGS)
 - Developmental and/or Epileptic Encephalopathy with Spike-Wave Activation in Sleep (DEE-SWAS, EE-SWAS)
 - Febrile Infection-Related Epilepsy Syndrome (FIRES)
 - Hemiconvulsion-Hemiplegia-Epilepsy Syndrome (HHE)

Mesial temporal lobe epilepsy with hippocampal sclerosis

- **Common cause of temporal lobe seizure**
 - Onset typically adolescent/ young adult
- **Cause**
 - Acquired abnormality in the hippocampus: loss of pyramidal neurons, granule cell dispersion and gliosis
 - Many contributing factors: genetic, genetic-structural, immune pathology
- **Seizures**
 - Focal temporal seizures (aware/ impaired awareness)
- **EEG**
 - Anterior temporal/ midtemporal epileptiform discharges
- **MRI**
 - Hippocampal sclerosis
- **Treatment**
 - Appropriate ASM
 - Often drug-resistant/ Refractory epilepsy

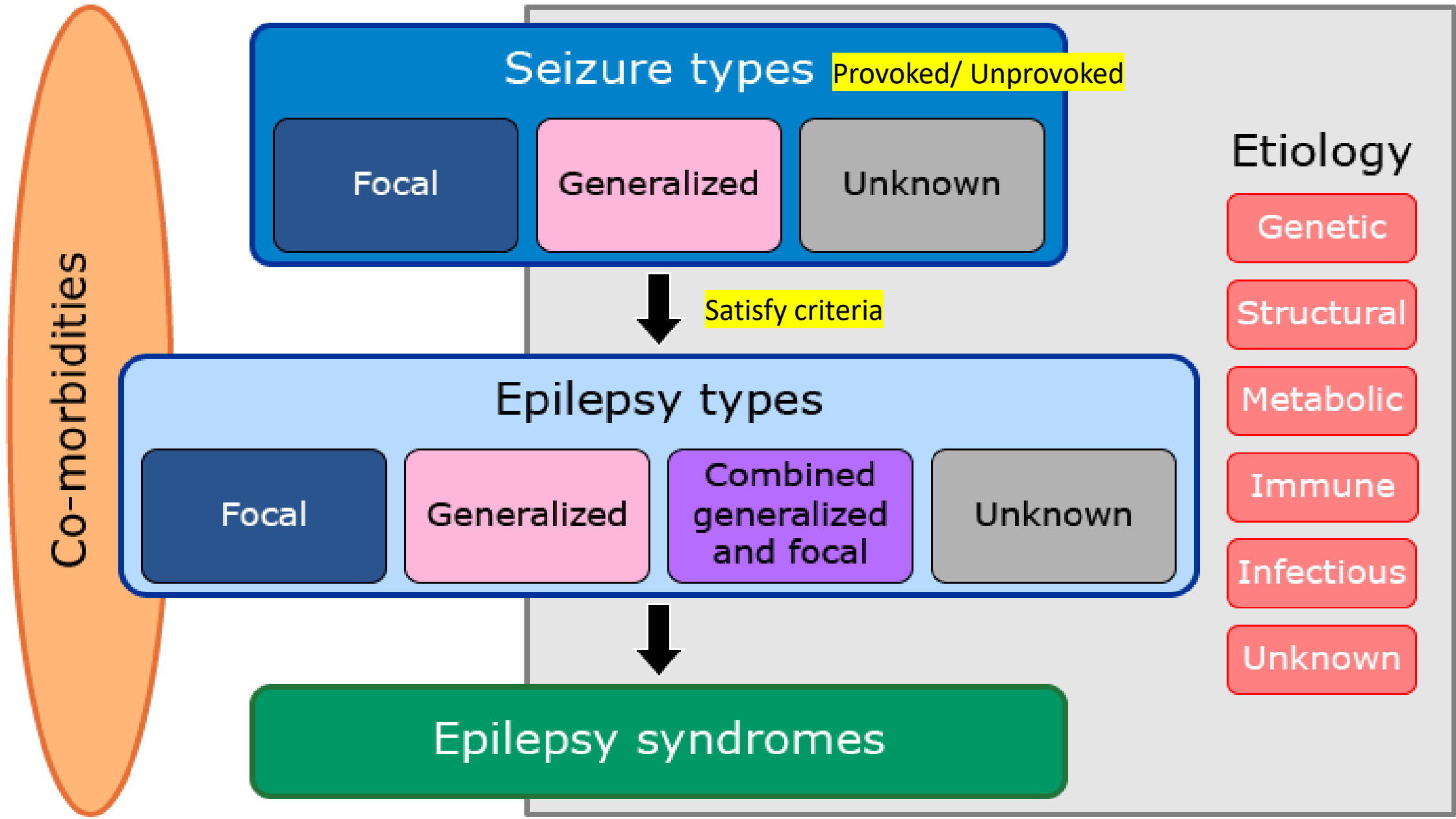
Juvenile myoclonic epilepsy

- **Common idiopathic generalized epilepsy syndrome**
 - Onset typically 10-24 years, female preponderance
 - Normal development and cognition
- **Cause**
 - Complex/ polygenetic
- **Seizures**
 - Myoclonic seizures within an hour of waking up
 - Subsequent development of GTCS
- **EEG**
 - 3-5.5Hz spike-wave discharge
- **Normal MRI**
- **Treatment**
 - Phenytoin/ carbamazepine may worsen myoclonus
 - Valproate acid treatment of choice (Lamotrigine/ Levetiracetam for women of child-bearing age)

Epilepsy of generalized tonic-clonic seizures alone

- **Previously known as epilepsy with grand mal seizure**
 - Onset typically 10-25 years
 - Normal development and cognition. Ppt by sleep deprivation/ alcohol
- **Cause**
 - Complex/ polygenetic
- **Seizures**
 - Generalized tonic-clonic seizure
- **EEG**
 - 3-5.5Hz spike-wave discharge
- **Normal MRI**
- **Treatment**
 - Valproate acid, Lamotrigine/ Levetiracetam for women of child-bearing age

International League Against Epilepsy (ILAE) framework for seizure and epilepsy classification



Clinical context

- A 66/M presenting with acute right frontal stroke and recurrent left arm involuntary movement
 - Acute symptomatic seizure due to stroke; not epilepsy
- A 25/M with recurrent epigastric sensation/ déjà vu followed by behavioral arrest. MRI brain showing hippocampal sclerosis
 - Focal epilepsy (temporal lobe epilepsy) due to hippocampal sclerosis
- A 18/F with a history of recurrent jerky movement in the morning, presenting with generalized tonic-clonic seizure. MRI normal.
 - Generalized epilepsy; epilepsy syndrome (Juvenile myoclonic epilepsy)
- A 82/F with known dementia presenting with generalized tonic-clonic seizure
 - Generalized epilepsy due to neurodegenerative disease

Further investigation

- Aid diagnosis/ risk stratification
 - EEG
- Look for cause
 - Blood test for electrolytes, glucose, kidney/liver function; further blood test like immune workup/ autoimmune encephalitis antibody panel/ toxicology screen as appropriate
 - CT/MRI brain
 - Lumbar puncture for CSF as appropriate

Electroencephalogram (EEG)

- To detect abnormal electrical activities (epileptiform discharges) in the brain
- The diagnosis of seizure/ epilepsy is almost always based on clinical history (usually from witness)
 - Around 50% of epilepsy patient has normal interictal EEG
 - Common to have false positive/ borderline positive EEG -> overdiagnosis epilepsy
 - Main use of EEG in epilepsy/ seizure
 - Support diagnosis of epilepsy
 - Risk stratification for initiation of antiseizure medication
 - Diagnosis of ongoing seizure/ non-convulsive status epilepticus
- ** You are not expected to know how to interpret EEG, but only know when to order one



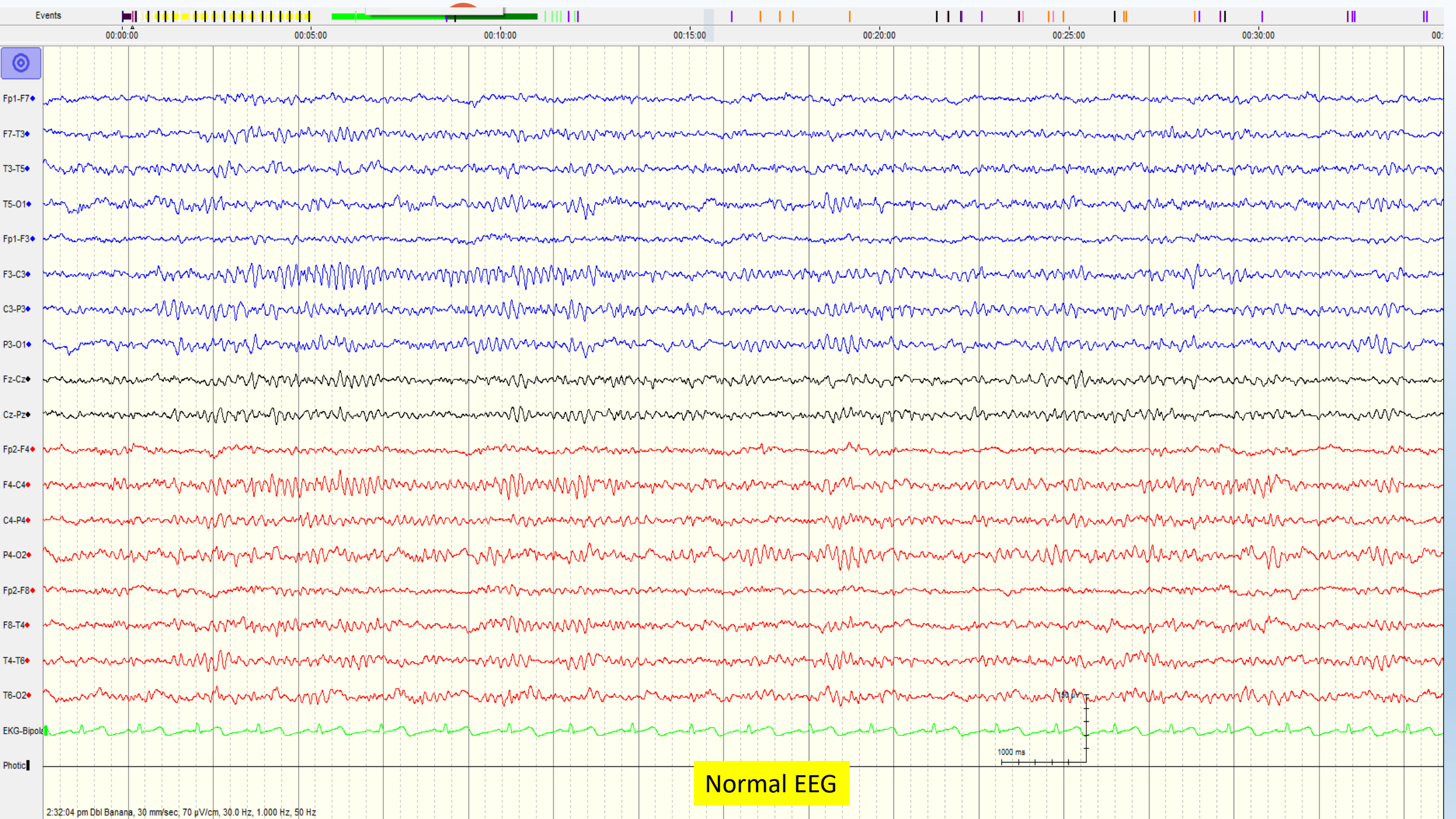
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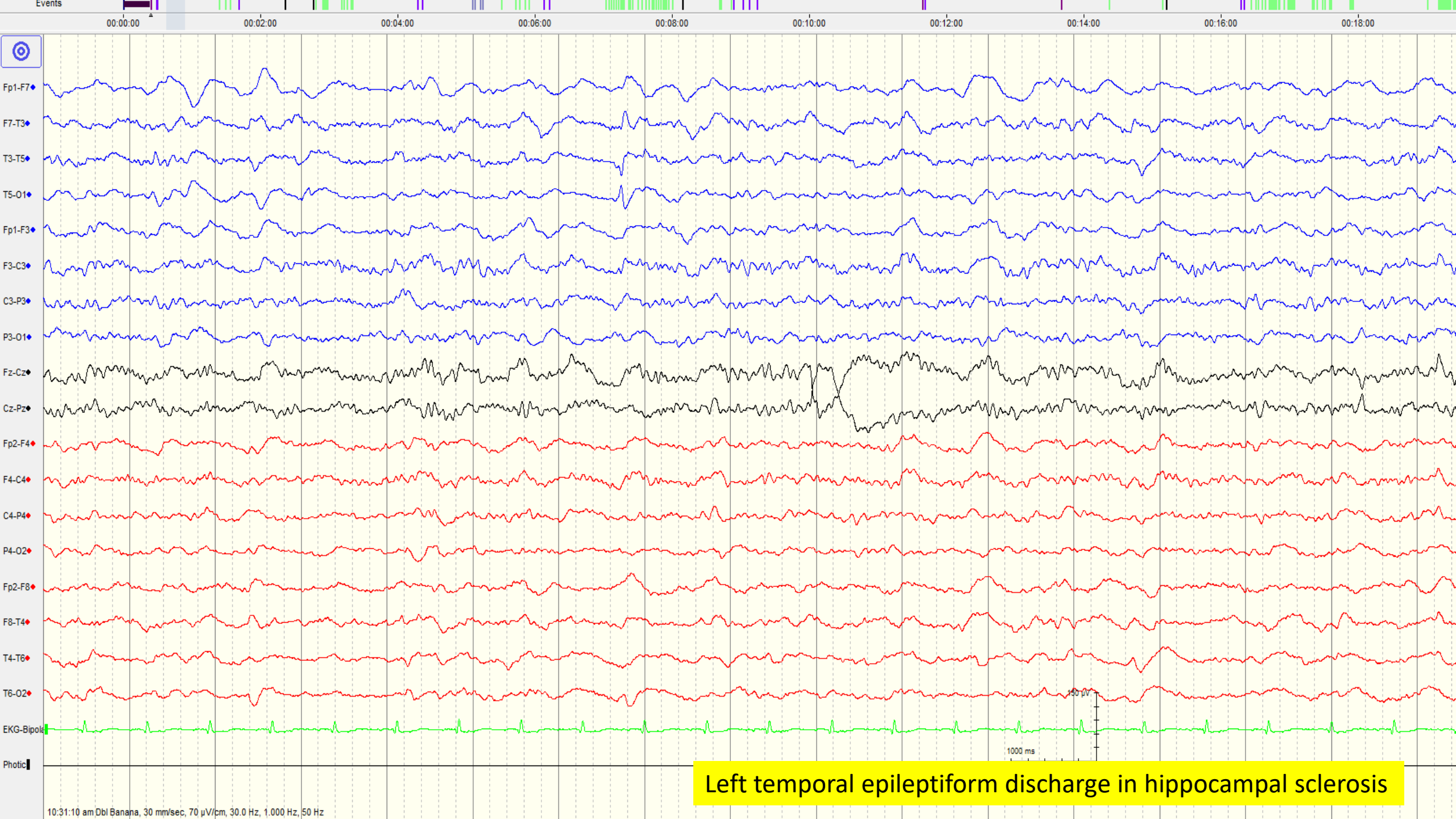
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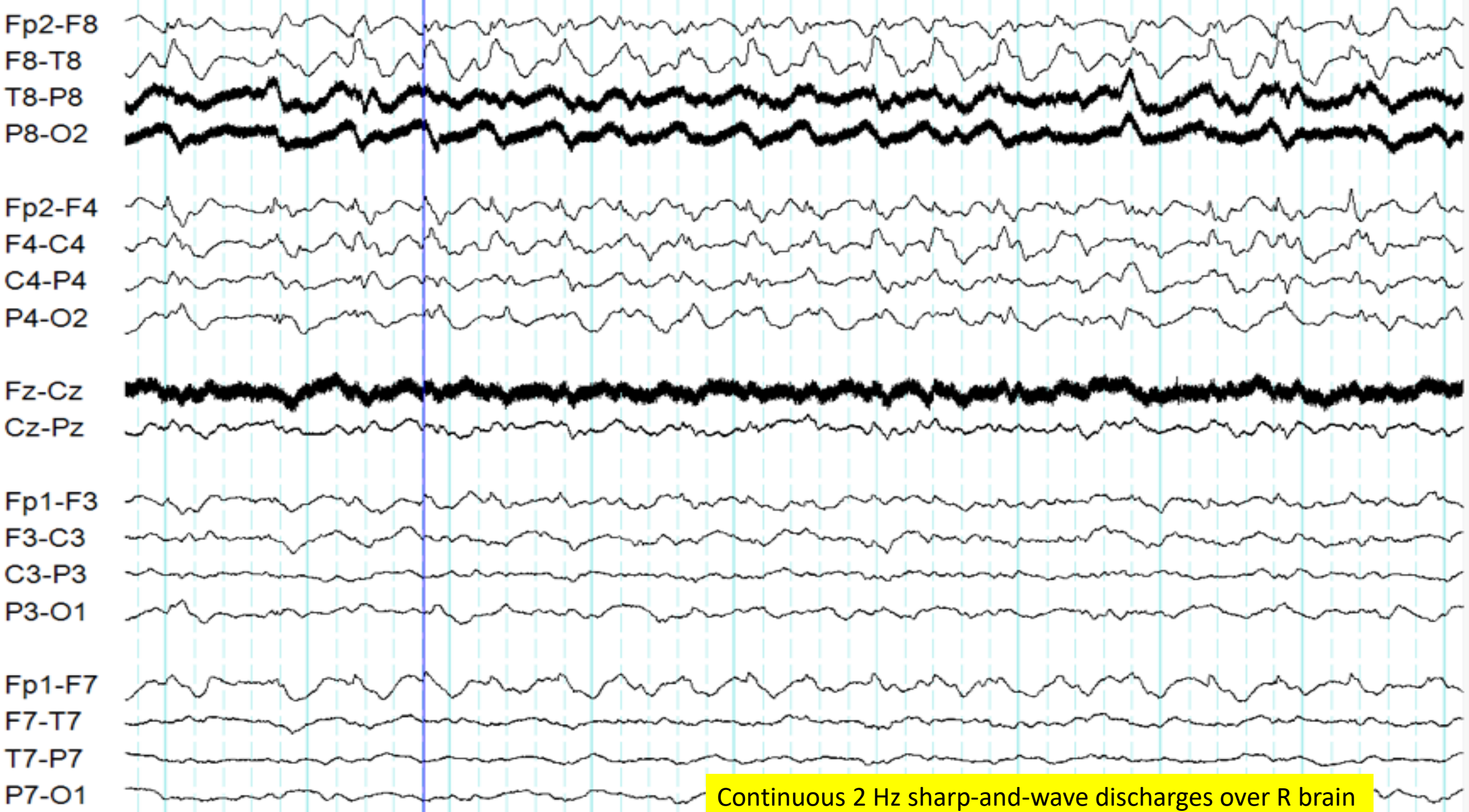


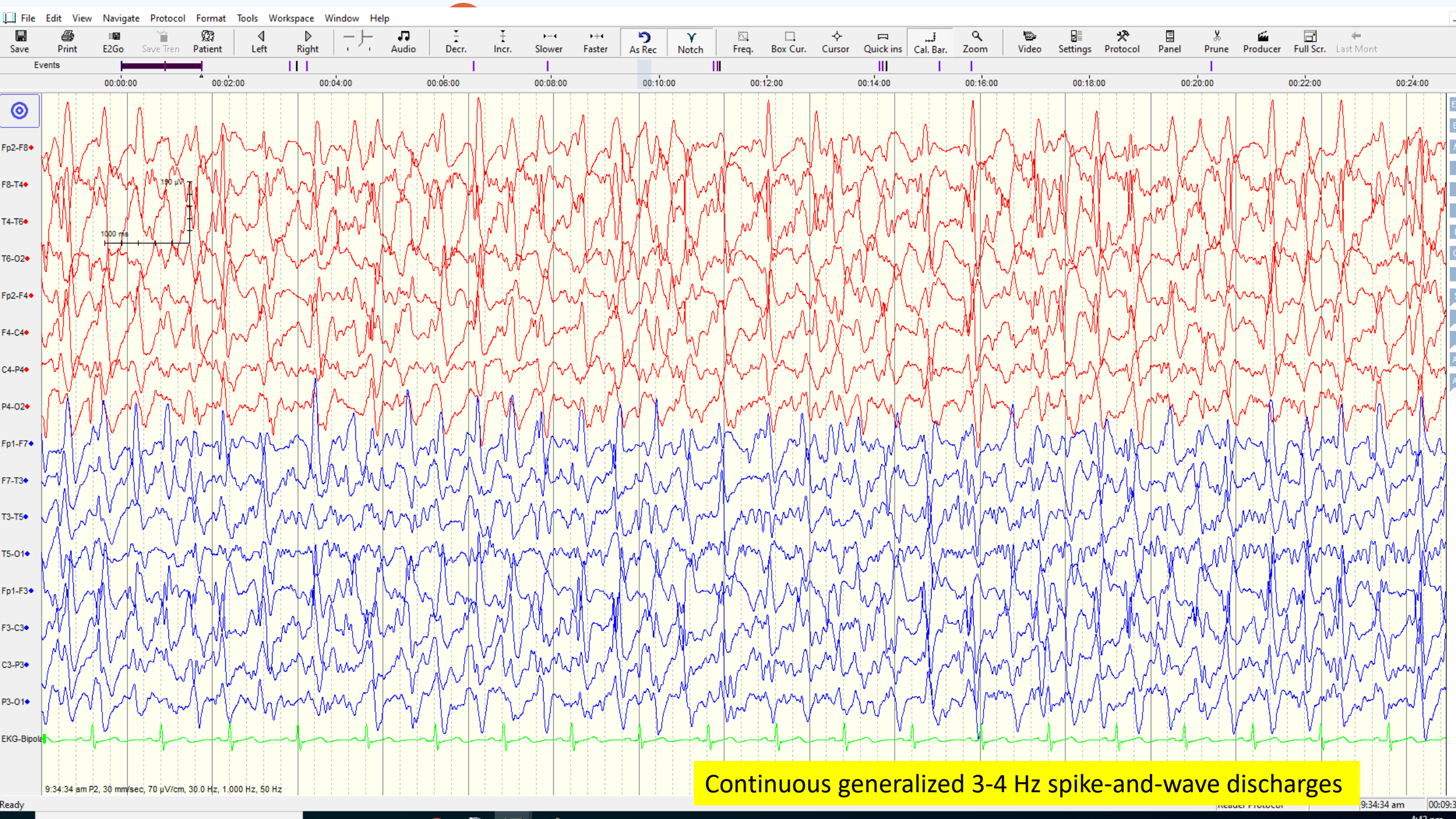


F7-AVG
T3-AVG
T5-AVG
Fp1-AVG
F3-AVG
C3-AVG
P3-AVG
O1-AVG
Fz-AVG
Cz-AVG
Pz-AVG
Fp2-AVG
F4-AVG
C4-AVG
P4-AVG
O2-AVG
F8-AVG
T4-AVG
T6-AVG



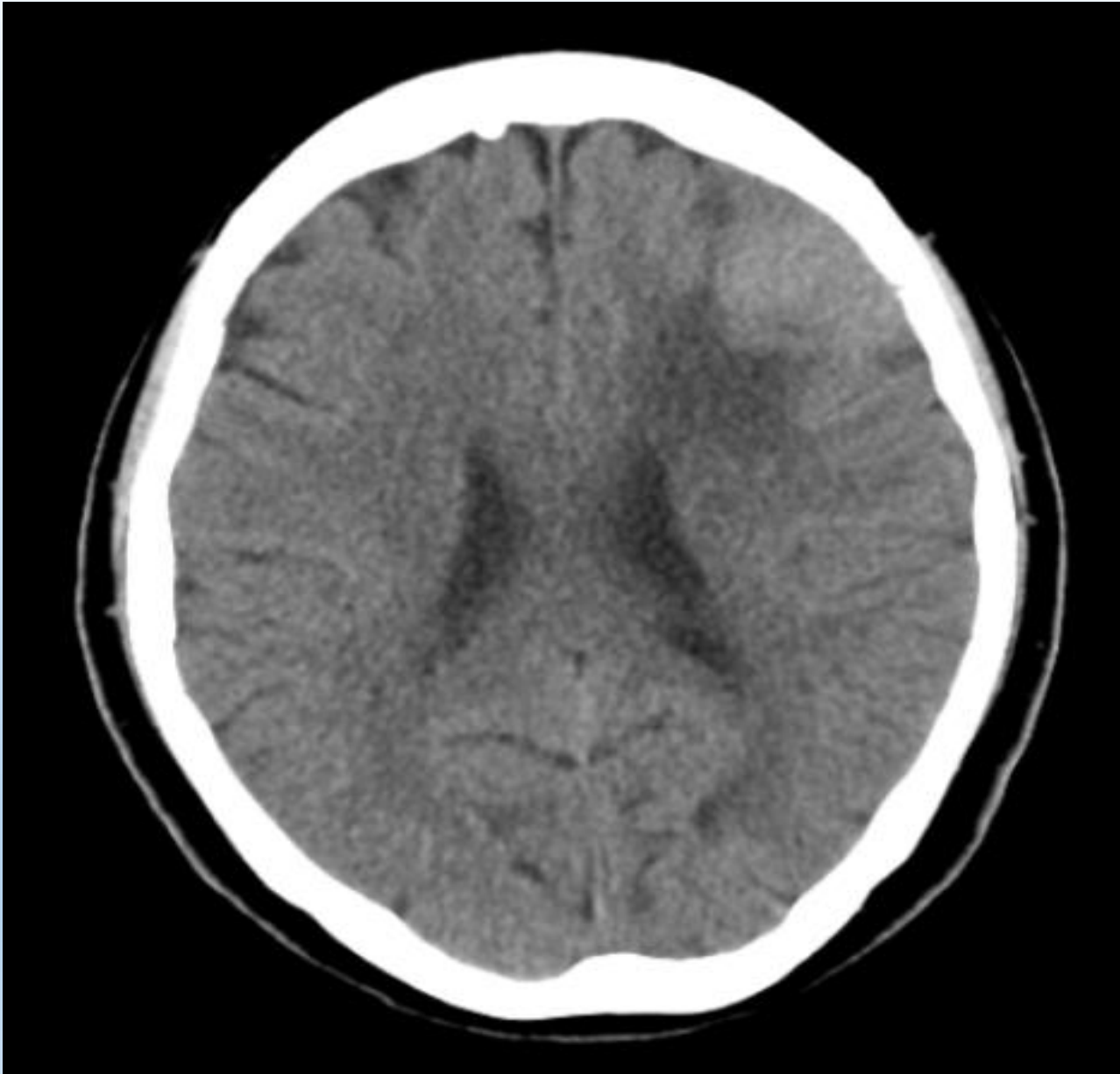
Generalized spike-and-wave discharge in JME



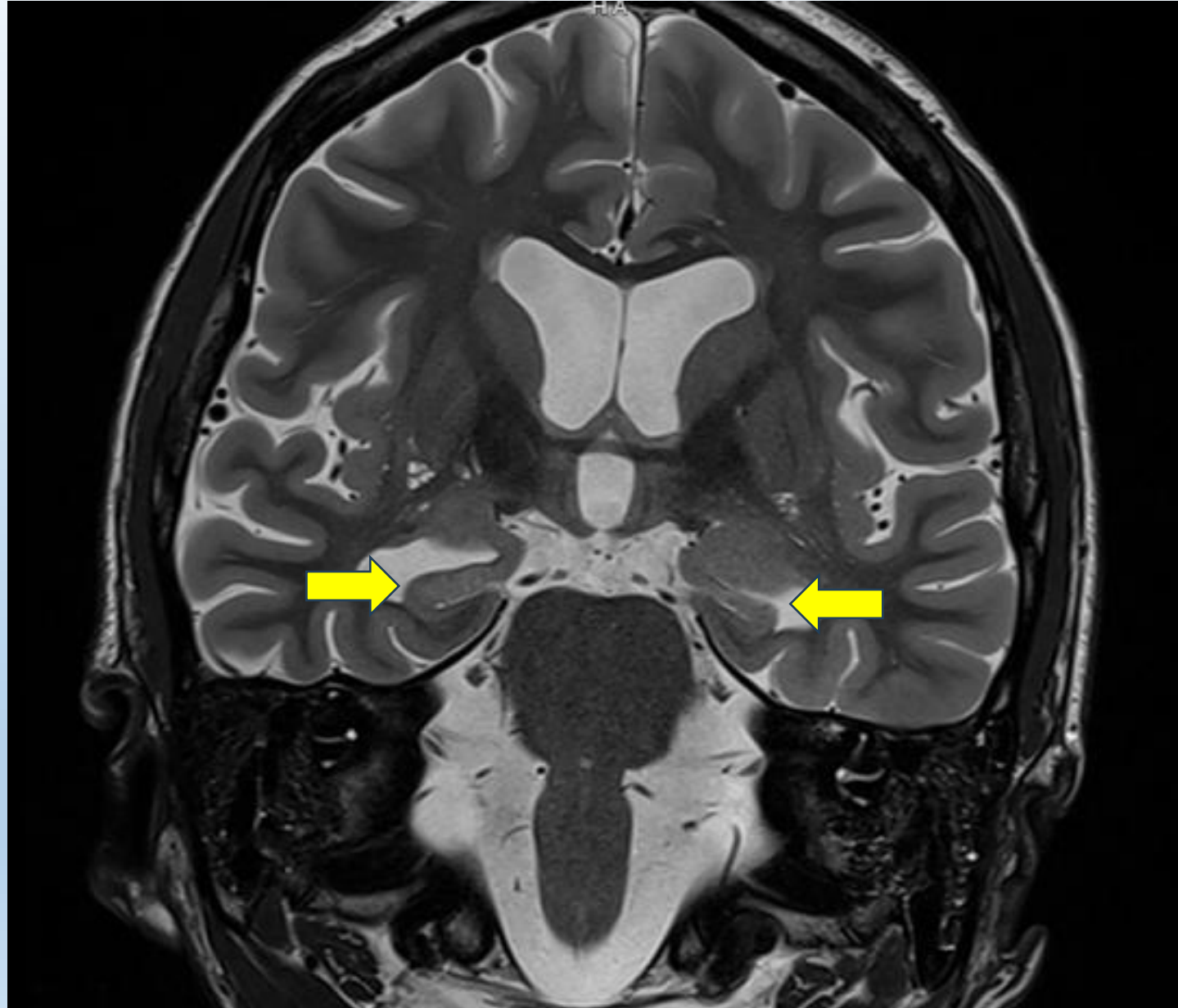


Neuroimaging

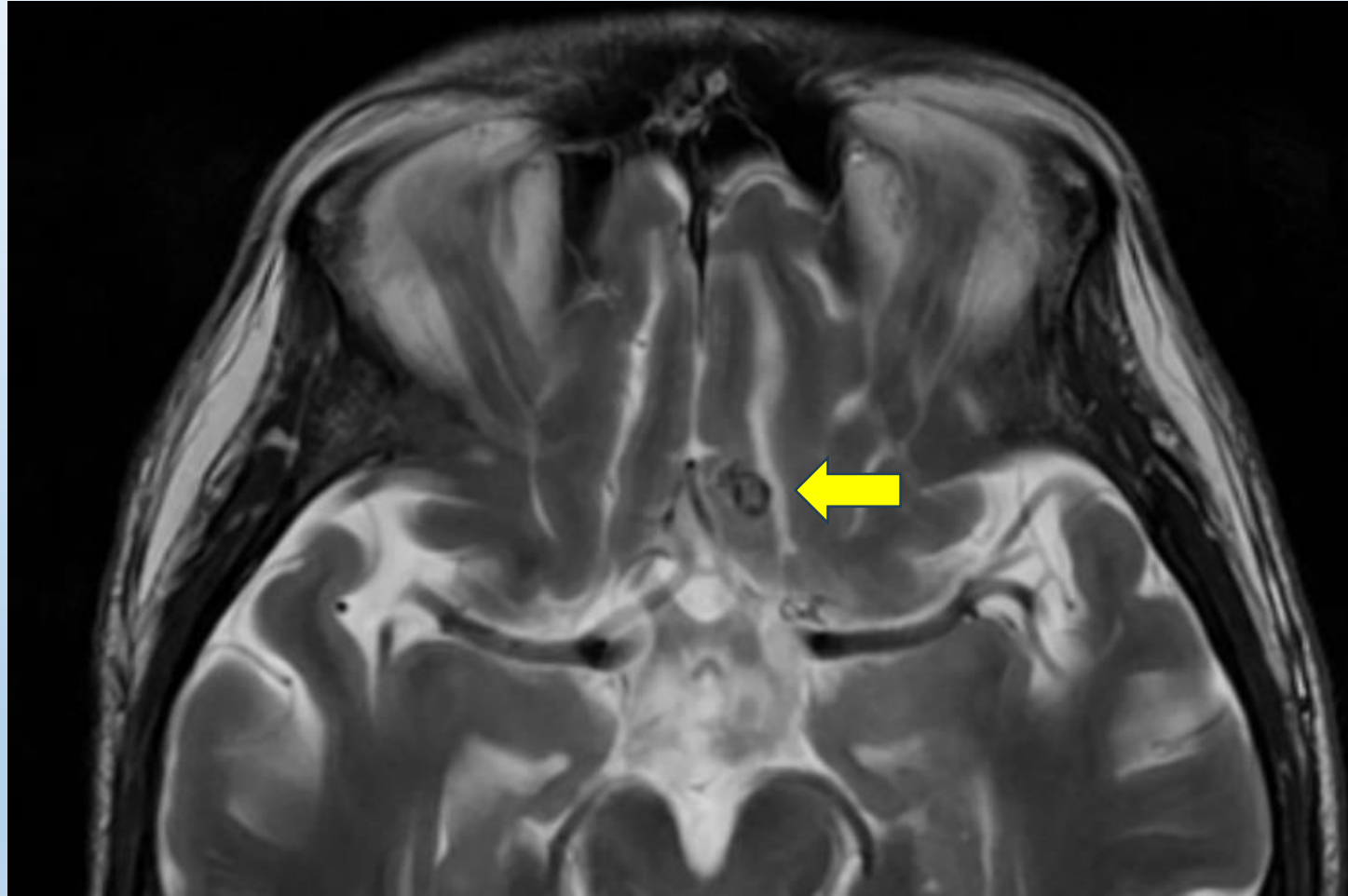
- Look for structural lesions
 - Brain imaging may be normal in many patients
- CT for obvious lesions
- MRI would probably be needed for all patients without obvious structural lesion on CT



Left frontal meningioma



Mesial temporal sclerosis



Left frontobasal cavernoma

Management of seizure/ epilepsy

- **First adult seizure**
 - Ensure full recovery (not in status epilepticus)
 - Rule out acute, potentially reversible causes (vascular, infection, inflammation, metabolic, toxic etc.)
 - **Make sure it's the first seizure** (patient may have recurrent behavioral arrest/ myoclonus that family/ patient is unaware is relevant)
- **Assess risk of recurrence (epilepsy diagnosis)**
 - EEG/ neuroimaging



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Management of seizure/ epilepsy

- General advice and seizure precautions
- Pharmacological Rx (antiseizure medications)
- Surgical Rx

General advice and seizure precautions

- **Diagnosis of epilepsy**
 - Considerable effect on mood, interpersonal relationship, employment, quality of life and ability to drive
 - Driving in HK -> report to transport department. No specific regulation to when patients with epilepsy can drive, hence, usually cannot drive indefinitely
- **Avoid activities that have ↑ risk of injury if seizure occurs**
 - Driving, operating high-risk equipment, working at heights
 - Swimming/ hiking alone
 - Cooking over fire alone
- **Risk of SUDEP (Sudden Unexpected Death in Epilepsy) ~1/1000**
 - Risk reduced with better control of epilepsy

Antiseizure medications (ASM)

- Used to be known as antiepileptic drug (AED)
- Drugs that suppress seizure by changing neurotransmission
 - **Sodium channel**
 - Carbamazepine, Lamotrigine, Phenytoin
 - **Calcium currents**
 - Ethosuximide
 - **GABA activity**
 - Benzodiazepines, phenobarbital
 - **Glutamate receptor**
 - Topiramate
 - **Synaptic vesicle proteins**
 - Levetiracetam
- Most acts through a primary pathway but also have effect on other pathways

Antiseizure medications (ASM)

- **Broad-spectrum (broad range of seizure: focal or generalized)**
 - Valproate
 - Levetiracetam
 - Lamotrigine
 - Topiramate
 - Clobazam
- **Narrow-spectrum (primarily focal onset, but not exclusive)**
 - Carbamazepine/ Oxcarbazepine
 - Phenytoin
 - Phenobarbitone
 - Lacosamide
 - Gabapentin/ pregabalin

Carbamazepine

- Block voltage-dependent sodium channel
- Effective for partial or generalized seizure, but can aggravate absence/ myoclonic seizures
- CYP enzyme inducer
 - Significant interaction with other ASMs, warfarin, DOAC, Rifampicin, Isoniazid, theophylline, prednisolone, haloperidol, OC pills
- Induce own metabolism -> need to increase dose after 1st month
- Hepatic metabolism
- Must check HLA B1502 before prescribing -> Those +ve: significant higher risk to develop SJS/TEN
- SE: Dizziness, somnolence, rash, **SJS, hyponatremia, cardiac arrhythmia**, marrow suppression, hepatitis

Phenytoin

- Block voltage-dependent sodium channel
- Effective for partial or generalized seizure, but can aggravate absence/ myoclonic seizures
- CYP enzyme inducer -> Drug interaction
- Metabolism through hepatic P450 mixed oxidase system -> enzyme saturable -> zero-order kinetics
 - Therapeutic window -> near enzyme saturation point -> small change in dose may lead to dramatic change in drug concentration
- Highly protein bound -> low serum protein can change concentration of unbound fraction
- Most common 1st line treatment for status epilepticus
- SE: Dizziness, somnolence, rash, dose-related neurotoxicity (nystagmus, unsteady gait), **gingival hyperplasia, coarsening facial features, hirsutism, hepatitis**, hypocalcemia (impair Vit D absorption), low folate

Valproate

- Effective for focal or generalized seizures
- Multiple cellular mechanism
 - Voltage-dependent sodium channel
 - GABA
 - Calcium currents
- Liver metabolism; CYP enzyme inhibitor
- Highly protein bound
- Teratogenic + children of pregnant ladies taking valproate -> lower IQ score -> Avoided in female of child-bearing age
- SE: Dizziness, somnolence, tremor, weight gain, hyperammonemia, hepatitis, pancreatitis, impair platelet function

Lamotrigine

- Effective for focal or generalized seizures
- Block voltage-dependent sodium channel
- Hepatic metabolism, excreted in urine
- Treatment of choice for female of child bearing age
- Few drug interaction
- Slow in action, gradual titration every 2 weeks -> reduce incidence of cutaneous S/E
- SE: **Rash (5-10%), SJS**, dizziness, somnolence, tremor, nausea
- Warn patient of rash, and stop ASAP and seek medical advice if rash occurs

Levetiracetam

- Act through synaptic vesicle proteins -> modulate exocytotic function
- Effective for focal or generalized seizures
- Partially metabolized by enzymatic hydrolysis, NOT dependent on P450 system
- Drug and metabolite excreted in urine
- Favored in patients with liver impairment
- Few drug interaction
- SE: Dizziness, somnolence, Mood/psychiatric (hostility, depression anxiety)

Choosing the correct ASM

- **Drug effectiveness**

- Focal or generalized epilepsy
- Childhood absence epilepsy: Ethosuximide

- **Potential adverse effects**

- Phenytoin: gingival hyperplasia, coarsening facial features, hirsutism, hepatitis
- Valproate: tremor, weight gain, hepatitis
- Levetiracetam: Avoid in patients with psychiatric illness

- **Childbearing plans**

- Avoid Valproate and Topiramate for pregnancy
- Lamotrigine and Levetiracetam preferred

Choosing the correct ASM

- **Interaction with other medications**
 - Enzyme inducer: Carbamazepine, phenytoin, phenobarbital
 - Enzyme inhibitor: Valproate
 - Important drug-drug interactions: Warfarin, DOAC, Rifampicin, Isoniazid, OC pills
- **Comorbid medical conditions esp. hepatic and renal disease**
 - Patient with liver impairment: Levetiracetam
 - Patient with renal impairment: dose adjustment of ASM
- **Patient preference**
- **Cost**

Response to antiseizure medications

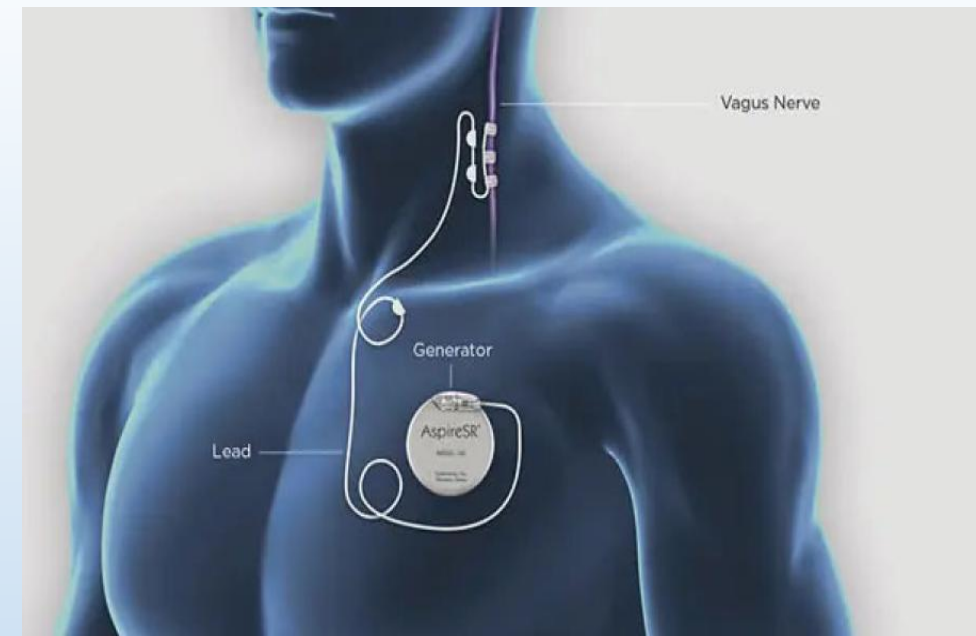
- Response to first ASM: ~50%
- Response to second ASM: additional ~ 15%
- Response to third ASM or multiple drug: additional ~5%
- Drug-resistant/ Refractory epilepsy
 - 1/3 of patients
 - Failed to become seizure free with adequate trial of two antiseizure medications (either monotherapy or combination)

Surgical management

- Epilepsy surgery (resection of area of the brain causing seizure)
- Highly selected patients with epileptic foci identified
- Extensive workup and surgical planning
 - Localization
 - Functional impact of the resection
- Classical example
 - Anterior temporal lobectomy or hippocampectomy for hippocampal sclerosis
 - Achieve 60-70% seizure freedom rates vs. 0-10% on medical Rx
 - Complications: surgical complications (infection, visual field defect, hemiparesis), memory impairment

Surgical management

- Vagal nerve stimulation therapy
- Wire to send impulse to vagus nerve
- Exact mechanism uncertain
 - Altered activities in the reticular activating system, central autonomic network, limbic system and noradrenergic projection network
- Reduce frequency and duration of seizure
- ‘Palliative’, in theory cannot achieve seizure freedom



Status epilepticus

- Definition
 - Continuous convulsive seizure for more than 5 minutes; or
 - Failed to regain consciousness after seizure
 - Non-convulsive status epilepticus (up to 60% of SE are non-convulsive)
 - Frequent, recurrent seizure close together
- **MEDICAL EMERGENCY**
- Recognition of status epilepticus is important, as prolonged seizure will lead to neuronal damage
- Aim to seizure abortion within the 1st hour. The longer the seizure, the more difficult the seizure can be aborted

Causes

- **Acute cerebral insult**
 - CNS infection, e.g. HSV encephalitis
 - Autoimmune encephalitis
 - Stroke
 - Brain tumour
 - Metabolic encephalopathy
- **Withdrawal/ Stop ASM**

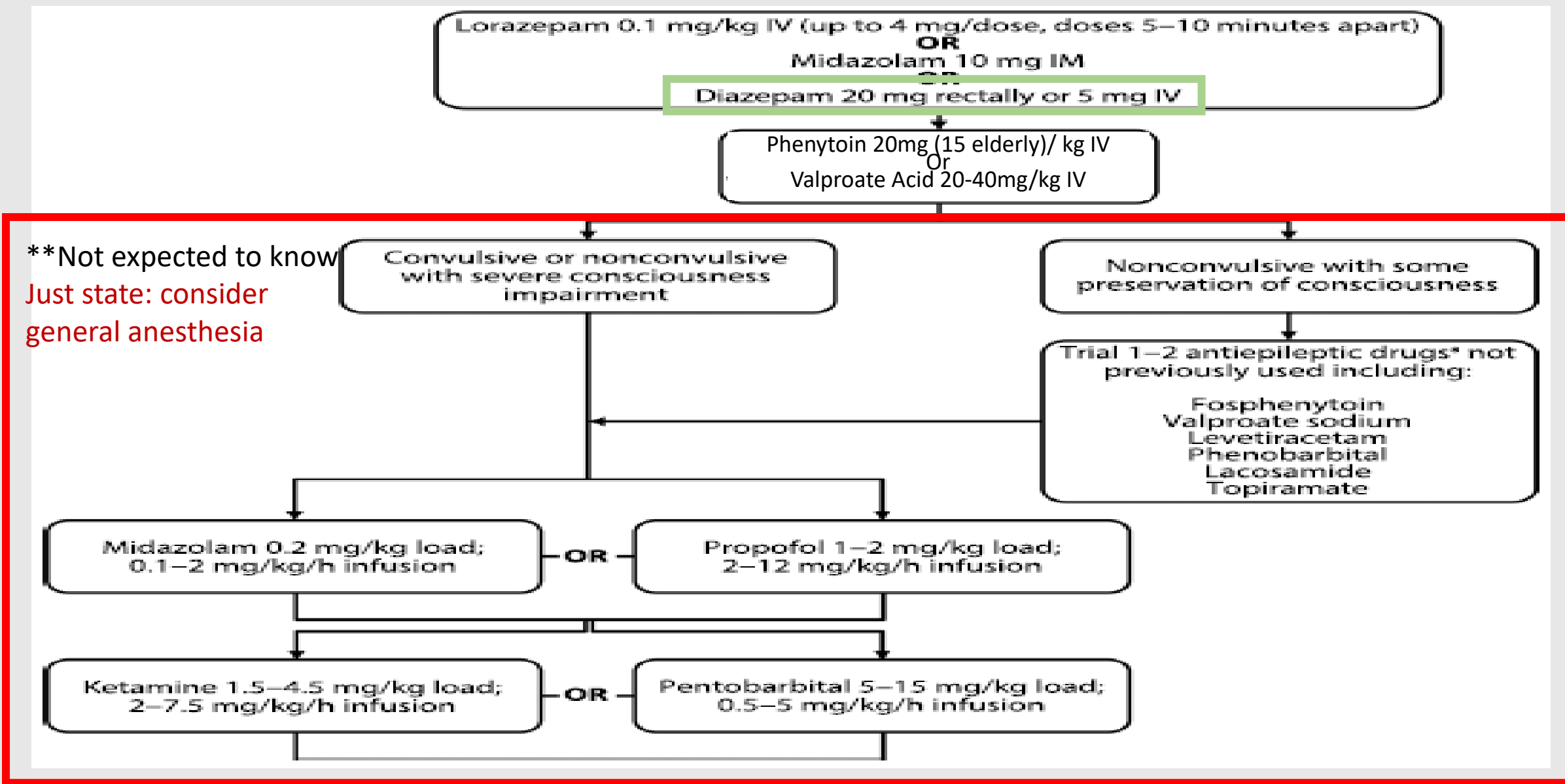


FIGURE 6-2

Suggested algorithm for management of status epilepticus.

IM = intramuscular; IV = intravenous; PE = phenytoin equivalent.

^a All drugs are IV administered except for topiramate, which is orally administered.

Management

- Close neuro-observation
 - Consider ICU admission
- Ensure ABC
- Prompt initiation of ASM
- Continuous EEG monitoring
- Workup and treat possible underlying cause
 - Tumour
 - Infection
 - CNS infection
 - Autoimmune encephalitis
 - Metabolic causes
- Prevent complications



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Brain death

Definition

- Brain death = brainstem death
 - Irreversible cessation of brainstem function
- Despite philosophical arguments, brain death = death legally in common law jurisdiction
 - After certified brain death, patient is medically certified dead
- Brain death is diagnosed
 - Severe, irreversible brain injury (patient in coma, and usually confirmed with neuroimaging)
 - Exclusion of potentially reversible causes
 - Absent brainstem reflexes

Procedure

- Specific criteria may differ by region/ country
- Diagnosed irreversible severe brain injury
 - **Waiting time after brain injury: at least 4 hours, 24 hours for CPR**
- Hong Kong (brain death certification)
 - **Done by 2 separate doctors**
 - One specialist of appropriate college (ICU, Neurologist, Neurosurgery)
 - Another at least 6 years after registration, but preferably be of same qualification as 1
 - **Preconditions**
 - Irreversible severe brain injury
 - Neuromuscular function is intact
 - Absence of depressant drugs
 - Absence of reversible causes contributing to a comatose state
 - Ensure normothermic, correct metabolic/ electrolyte disturbance, stable hemodynamic
 - **Proceed to clinical testing of brainstem reflexes**

Clinical test

- **Brain stem reflexes** (ALL test MUST be absent response)
 - No pupillary responses
 - No corneal reflex
 - No vestibular-ocular reflex (cold caloric test)
 - Absent of motor response on painful stimulation within trigeminal distribution
 - No gag and cough reflex
 - No respiratory movement with apnea test
- **Confirmatory test**
 - Only done when 1) no clear cause of coma; 2) possible metabolic/ drug effect; 3) any of the brainstem test cannot be completed
 - **Neuroimaging:** Brain perfusion imaging or four-vessel DSA to demonstrate absent brain perfusion/ flow

Summary

- **Consciousness and it's disorders**

- Describe the differential diagnosis and management of altered state of consciousness.
- Perform the Glasgow Coma Scale (GCS).
- Identify features that differentiate syncope and seizure.

- **Seizure and epilepsy**

- Name the different classifications of seizure.
- Explain the definition of seizure and epilepsy.
- Describe the epidemiology, etiologies, and clinical features of different types of seizure/ epilepsy.
- Describe commonly used antiseizure medications and their side effects.
- Learn the emergency management of a seizure attack and status epilepticus.

- **Electrophysiology I**

- Recognize the value of EEG as an investigative tool in seizure/ epilepsy.

- **Brain death**

- Explain what is brain death and how it is diagnosed.